

Technical Document #160: Blockchain - Comprehensive Overview

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Blockchain is a distributed ledger technology that records transactions across many computers. It ensures transparency and immutability.

Proof of Work (PoW) is a consensus mechanism where miners solve complex puzzles to validate transactions. It is energy-intensive but secure.

Zero-knowledge proofs allow one party to prove knowledge of information without revealing the information itself. They enhance privacy on blockchains.

Tokenomics studies the economic systems of cryptocurrency tokens. It covers supply mechanisms, incentive structures, and value capture.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Key Takeaways:

- Proof of Stake (PoS) selects validators based on their stake in the network.
- Blockchain is a distributed ledger technology that records transactions across many computers.
- Decentralized finance (DeFi) recreates traditional financial systems using blockchain technology.